



Arnav Agarwal
Electrical Engineering
Indian Institute of Technology Bombay
Specialization: Communication & Signal Processing

22B3917
Dual Degree (B.Tech. + M.Tech.)
Gender: Male
DOB: 05/02/2004

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2027	8.63 Core: 204 Total: 267

Pursuing dual minors in **AI and Data Science** (C-MInDS) and **Management** (SJMSOM), IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Recipient of **KVPY** fellowship granted to **1800+** students out of **97k+** candidates by the Govt. of India (May'22)
- Recipient of the **IPM** scholarship (top 300 in India) with four-time Mega Finals distinction (top 500) (Jul'16)
- Awarded the **Best Project** in Electronics Design Lab for 3-DOF Robotic Arm among **200+** students (Apr'25)

PUBLICATIONS

- **A. Agarwal***, D. Sawke* et al., "Hardware-Aware Semantic Segmentation Engine for On-Chip Retinal Prosthesis Navigation," IEEE International Conference on Artificial Intelligence Circuits and Systems (**AICAS**) 2026
- **A. Agarwal***, D. Sawke* et al., "On-Chip Streaming Fixed-Point Vision Processing Pipeline for Epiretinal Prostheses," accepted for **lecture presentation** at the IEEE Asia Pacific Conference on Circuits and Systems (**APCCAS**) 2026

*Equal contribution

PROFESSIONAL EXPERIENCE

AI Researcher | IBM Research India (May'26 - Aug'26)

Awarded **Letter of Recommendation** from Senior Research Engineer, recognizing contributions to AI research team

- Designed a moment-field encoder+region-based **MoE** decoder to infer stochastic differential equations from trajectories
- Beat **FIM-SDE**: transformer foundation-model baseline; cut RMSE **27%** with **4×** fewer params, **4×** faster training
- Trained both models from scratch on a **400K**-system dataset in **PyTorch** on IBM's multi-GPU **HPC** cluster
- Paper on this work to be submitted to the **NeurIPS 2026** Workshop on AI for Stochastic Dynamics (STODY)

Machine Learning Engineer | GoComet India Private Limited (Feb'25 - May'25)

- Designed an **NLP** pipeline over **100k+** monthly news and RSS feeds, flagging port-risk events at **85%+** precision
- Trained the **PyTorch** classifier and deployed it on **AWS** via **FastAPI**, serving the predictions at under **2s** latency

Generative AI Engineer | Trupeer Technologies Private Limited (May'24 - Jul'24)

- Built **Golang** APIs for a Chrome extension generating manuals+AI videos in **<5 min** improving speed by **80%**
- Developed **NextJS** app for PDF extraction with **Tesseract** and multilingual TTS via Bhashini API (**<6s** latency)
- Selected to present at **UNESCO's** 46th World Heritage Committee Meeting among **2000+** delegates in India

RESEARCH EXPERIENCE

Generative AI Research Intern | Arizona State University, USA (May'25 - Jul'25)

A 2-month **on-site** internship under Prof. Ashif Iquebal in **generative 3D CAD modeling**

- Trained a generative **PyTorch** model on **CUDA** that maps 2D images to 3D **B-rep** CAD, BLEU-4 score of **0.72**
- Processed the raw **DeepCAD** dataset of **500k+** CAD models into B-rep and rendered images for multimodal training
- Trained autoregressive cascaded **Transformers** with a **Pointer Network** on multimodal B-rep and image dataset
- Authored a B-rep construction engine in **pythonocc-core**, exporting STEP files at **94%** validity on **5k+** models

On-Chip Semantic Segmentation for Retinal Prostheses | Thesis Project (Oct'25 - Present)

Guide: Prof. Laxmeesha Somappa, Department of Electrical Engineering

- Designing an **INT8 ASIC** vision core for a retinal-prosthesis chip worn on a blind user's spectacles, under a sub-**30KB** on-chip memory budget, **<100ms** latency, and fixed-point-only arithmetic, forcing a from-scratch compact design
- Designed and implemented a **7-stage INT8** streaming front-end (denoising, gamma correction, contrast enhancement, edge/saliency fusion) converting **640×480** spectacle-camera frames to **80×60**, entirely on-chip with no frame buffer
- Trained a custom **12,880-param U-Net** across **7** design iterations on **SANPO** for 4-class outdoor segmentation (background, traversable, obstacle, hazard), running entirely on-chip at **35.97** mIoU/M params, **21×** prior models

KEY PROJECTS

End-to-End Image Captioning for Remote Sensing | EE782: Course Project (Jul'25 - Aug'25)

Guide: Prof. Amit Sethi | Advanced Topics in Machine Learning, EE

- Designed a from-scratch captioning pipeline on **RSICD** (**10.9k** images) with a custom tokenizer and cached features
- Benchmarked four encoder-decoder setups in **PyTorch**, **ResNet-18** and **MobileNetV2** with LSTM and Transformer
- Scored BLEU-4 **0.2513** and METEOR **0.4604** with MobileNet+Transformer; GradCAM traced blur and vocab gaps

CUDA Kernels and LLM Inference Systems | CS794: Course Project (Jan'26 - Apr'26)

Guide: Prof. Mythili Vutukuru | Systems for Machine Learning, CSE

- Implemented **KV caching** and prefix-sharing in nanoGPT using a shared **radix tree** to reuse cached attention
- Optimized **CUDA** matmul naive to coalesced, **6.8×** faster on **4096×4096** matrices vs **cuBLAS** at 3.5 TFLOPS
- Developed a 2-layer **MLP** forward and backward in raw CUDA with CUDA streams and gradient checkpointing
- Implemented **FlashAttention** from scratch in CUDA with online-softmax tiling, verified against PyTorch's kernel

Generative Models for Few-Shot Medical Imaging | EE782: Course Project (Sep'25 - Nov'25)

Guide: Prof. Amit Sethi | Advanced Topics in Machine Learning, EE

- Built a few-shot benchmark on **HAM10000** (**7-class** dermoscopic skin-lesion dataset), training a **ResNet-18** classifier in **PyTorch** on **350 real images** (50/class) with **Adam** optimization to evaluate generative augmentation strategies
- Trained three class-conditioned generative models from scratch: **CVAE**, **DCGAN**, and a **U-Net DDPM diffusion** model, with one-hot label conditioning to synthesize **1,200** dermoscopic images across the **6** minority classes
- Augmented training with real+synthetic mixes boosting accuracy **+6.6** (diffusion), **+4.1** (CVAE), **+1.9** (GAN) pts

Real-Time ADAS Semantic Segmentation on FPGA | EE712: Course Project (Jan'26 - Apr'26)

Guide: Prof. Laxmeesha Somappa | Embedded Systems Design, EE

- Trained a custom **SimpleFCN** (**11K** params) in **PyTorch** for real-time 4-class road-scene segmentation at **128×128**
- Applied **INT8** quantization, reducing FP32 weight memory by **4×** to fit a tight **630KB** on-chip inference budget
- Deployed the INT8 model via **Vitis HLS** as custom IP on **PYNQ-Z2** (XC7Z020: 87% BRAM, 17% LUT, 13% DSP)

Automated Medical Coding | Summer Project, IIT Bombay (Mar'24 - May'24)

Guide: Dr. Kshitij Jadhav, Koita Centre for Digital Health, IIT Bombay

- Designed a hierarchical **ICD-9** pipeline on **MIMIC-III** discharge notes, splitting by specialty via stratified sampling
- Fine-tuned **Mistral-7B** in two stages on ICD-9 code descriptions and clinical notes for multi-label ICD assignment

Deep Neural Network Robustness | CS781: Course Project (Aug'24 - Oct'24)

Guide: Prof. Supratik Chakraborty | Formal Methods in Machine Learning, CSE

- Certified a CIFAR-10 ResNet with α , β -**CROWN** and **auto_LIRPA** under $L_\infty \epsilon = 2/255$ and **5×5** spatial patches
- Applied α -**CROWN** bounds and β -**CROWN** branch-and-bound after **PGD** filtering of non-robust images on GPU

Classical ML for Vehicle Motion Classification | DS203: Course Project (Jan'26 - Apr'26)

Guide: Prof. Vinay Kulkarni | Programming for Data Science, C-MInDS

- Performed **EDA** on class-wise motion and engineered **23** hand-crafted features: optical flow, Canny edges, region diffs
- Trained **Random Forest**, Gradient Boosting, Support Vector Machine, **XGBoost** and Logistic Regression with video-wise train/test splits, class and sample weights, and **GridSearchCV** under GroupKFold cross-validation
- Reached **95%** accuracy and **0.86** macro-F1 after GroupKFold tuning and temporal smoothing on held-out videos

INTERNATIONAL EXPOSURE

Entrepreneurship Bootcamp 2026 | HKUST, Hong Kong (Jul'26)

India's sole **AUA-HKUST** delegation: one of **3** IIT Bombay nominees for a **7-day on-site** bootcamp

- Worked with founders and startup mentors on design thinking and pitching during the 7-day on-site bootcamp
- Won **3rd prize** among 15+ teams at HKUST for pitching **Buddy Booth**, an AI bully-detection kiosk in schools

POSITIONS OF RESPONSIBILITY

Teaching Assistant | EE340: Communications Lab (Aug'26 - Present)

- Coordinating weekly GNURadio SDR and hardware labs, mentoring and guiding **12** students through lab assessments
- Assisting the instructor in running labs, evaluating submissions, and handling student queries during the semester

Teaching Fellow | SATHEE Project, IIT Kanpur, Ministry of Education (Apr'23 - Present)

- Conducting **10+** teaching sessions to guide students in IIT JEE preparation within the SATHEE Project
- Mentoring JEE aspirants on Physics problem-solving, supporting concept review and exam-oriented practice

TECHNICAL SKILLS & COURSES

Programming	Python, C, C++, CUDA, Go, Embedded C, JavaScript, Assembly, TypeScript, SQL, Bash
Development	React, NextJS, NodeJS, Express, FastAPI, HTML, CSS, Docker, Jupyter, Git, Linux, AWS
Libraries	PyTorch, TensorFlow, HuggingFace, LangChain, NumPy, Pandas, Scikit-Learn, OpenCV
Hardware	VHDL, SystemVerilog, MATLAB, Cadence, Quartus, Keil, NGSpice, Arduino, TCL
AI/ML	Adv. Topics in ML, DL for NLP, Systems for ML, Formal Methods in ML, Speech and NLP

EXTRACURRICULARS

- Won **2nd prize** at the Searce × Google Cloud GenAI Hackathon, youngest among 20 industry teams (Feb'24)
- Won **1st prize** at ENPRO (E-Cell, IIT Bombay) for cutting empty-miles **65%** and logistics cost **45%** (Aug'23)
- Selected as one of **20** delegates across India to participate in the Chhatra Sansad InternNation Tour 2025 (Jan'25)
- Participated **twice** in the school band for sports day, as both a **bugle player** and **drummer** (Jul'19)